Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Winter 2019 — Problem 1

Row reduce $[\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 2 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

(a) A pivot in every column, so the set is **linearly independent**.

(b) Three independent vectors: $\boxed{\dim = 3}$.

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Math 208 Final, Winter 2019 — Problem 2

(a) $A\mathbf{x} = \begin{bmatrix} -10+9+0+1 \\ -5+3+1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓.

(b) $A \sim \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & -1 \end{bmatrix}$, so nullity $= 2$ with basis $(-3,2,1,0), (-2,1,0,1)$ (note $\mathbf{x}$ is their sum). Since $\dim \operatorname{null}(A) = 2$, any two independent null vectors form a basis:

$$\boxed{\{\mathbf{x},\ (-3,2,1,0)\}} \quad \text{(not multiples of each other).}$$

Math 208 Final, Winter 2019 — Problem 3

(a) $\mathbf{x} = 2\begin{bmatrix}2\\1\end{bmatrix} - 3\begin{bmatrix}1\\1\end{bmatrix} = \boxed{\begin{bmatrix}1\\-1\end{bmatrix}}$.

(b) Solve $a(2,1) + b(1,1) = (4,3)$: $a = 1$, $b = 2$, so $\boxed{[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix}1\\2\end{bmatrix}}$.

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Math 208 Final, Winter 2019 — Problem 4

Multiply on the right by $D + BX$: $AX = C(D + BX) = CD + CBX$. Then $(A - CB)X = CD$, so

$$\boxed{X = (A - CB)^{-1}CD}.$$

Math 208 Final, Winter 2019 — Problem 5

$A\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix} = \begin{bmatrix}1 & 2\\ 4 & -1\end{bmatrix}$, so

$$A = \begin{bmatrix}1 & 2\\ 4 & -1\end{bmatrix}\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}^{-1} = \boxed{\begin{bmatrix}1 & 0\\ -2 & 3\end{bmatrix}} \quad (\text{check: } A(1,2)=(1,4),\ A(2,1)=(2,-1)).$$

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Math 208 Final, Winter 2019 — Problem 6

Row reduce $[\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{v}_1\ \mathbf{v}_2] \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. So $\mathbf{v}_1 = \mathbf{u}_1 - 2\mathbf{u}_2$ and $\mathbf{v}_2 = 2\mathbf{u}_1 + 3\mathbf{u}_2$, giving $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \subseteq \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$. The coefficient matrix $\begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$ has determinant $7 \neq 0$, so $\mathbf{v}_1, \mathbf{v}_2$ are independent and span a 2-dimensional subspace inside a 2-dimensional one. **Yes, the spans are equal.**

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Math 208 Final, Winter 2019 — Problem 7

$\det(A-\lambda I) = (1-\lambda)(3-\lambda)+2 = \lambda^2 - 4\lambda + 5$, so $\boxed{\lambda = 2 \pm i}$.

For $\lambda = 2+i$: $(A-\lambda I)\mathbf{v} = \mathbf{0}$ with first row $(-1-i)v_1 - v_2 = 0$, so $\mathbf{v} = \begin{bmatrix} 1 \\ -1-i \end{bmatrix}$. For $\lambda = 2-i$ take the conjugate $\begin{bmatrix} 1 \\ -1+i \end{bmatrix}$ (any nonzero multiples also work).

Math 208 Final, Winter 2019 — Problem 8

(a) $\det(A-\lambda I) = (2-\lambda)\big[(-\lambda)(-1-\lambda)-6\big] = (2-\lambda)(\lambda+3)(\lambda-2)$: $\boxed{\lambda = 2 \text{ (twice)},\ -3}$.

$\lambda = 2$: $A - 2I \sim \begin{bmatrix} 0 & 1 & -\frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, basis $(1,0,0), (0,3,2)$. $\lambda = -3$: basis $(1,-1,1)$.

(b) **Yes** (three independent eigenvectors):

$$A = PDP^{-1}, \quad P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & -1 \\ 0 & 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Winter 2019 — Problem 9

(a) The spanning set contains $(1,2)$ and $(-1,2)$, which are independent, so $\operatorname{col}(A) = \mathbb{R}^2$ and we need rank 2: e.g. $\boxed{A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}}$ (nullity $4 - 2 = 2$).

(b) For example $\boxed{A = \operatorname{diag}(1,1,1,0)}$.

(c) **Not possible**: the kernel of a map $\mathbb{R}^4 \to \mathbb{R}^3$ has dimension at least $4 - 3 = 1$.

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Winter 2019 — Problem 10

(a) **True**: $\det(A^2) = (\det A)^2 = 0$. (b) **True**: 0 is not an eigenvalue ($\det A = 4 \cdot (-1)^2 = 4 \neq 0$). (c) **False**: e.g. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ has row space $\text{span}\{(1,0)\}$ but column space $\text{span}\{(1,1)\}$. (d) **False**: nonreal eigenvalues of a real matrix come in conjugate pairs, so $2+i$ would require $2-i$ too. (e) **False**: the system could be inconsistent.